04 - Text Model Optimisation

Goal: find the best settings for the text model.

1. Grid search with 3-fold cross-validation on the training set.
2. Compare LinearSVC with two other models.
3. Score the final model once on the validation set.

Validation is not used to choose any setting.

1. Load the data

```
Train:      67,932
Validation: 16,984
```

2. Grid search on LinearSVC

The 03 notebook showed that using all features beats a 50,000 limit by 1.2 points (CV on train). So we keep all features.

We test two settings together:

- `binary` : count how often a word appears (False), or only whether it appears (True)
- `C` : how closely the SVM fits the training data. A small C gives a simpler model.

That gives 8 combinations, each scored with 3-fold CV on train.

```
Grid search time: 1.8 min
```

	binary	C	cv_weighted_f1	cv_std
0	True	1.0	0.8284	0.0026
1	True	2.0	0.8282	0.0020
2	True	0.5	0.8263	0.0030
3	False	1.0	0.8235	0.0025
4	False	2.0	0.8231	0.0024
5	False	0.5	0.8223	0.0025
6	True	0.1	0.8031	0.0013
7	False	0.1	0.7999	0.0017

```
Binary gain:  +0.0050
CV spread:    0.0030
Use binary:   True
Best C:       1.0
```

Analysis

- C = 1 gives the best score. C = 0.5 and 2 are almost the same. C = 0.1 is about 2.5 points lower.
- Binary scores 0.8284 against 0.8235 without it. The gain (0.5 points) is larger than the CV spread (0.3), so we keep `binary=True`.
- Chosen settings: all features, `binary=True`, `C=1`.

3. Other models

We keep the chosen TF-IDF settings and only change the classifier:

- Complement Naive Bayes: a fast and simple text model.
- Logistic Regression (trained with SGD): a linear model like the SVM that also gives probabilities.

Each model is scored with the same 3-fold CV on train.

	model	cv_weighted_f1	cv_std	time_min
0	LinearSVC (chosen)	0.8284	0.0026	0.2740
1	Logistic Regression (SGD)	0.8084	0.0015	0.1829
2	Complement Naive Bayes	0.7600	0.0003	0.1147

Analysis

- LinearSVC has the best score (0.8284).
- Logistic Regression is 2 points lower (0.8084).
- Complement Naive Bayes is 7 points lower (0.7600).
- We keep LinearSVC.

4. Final score on validation

We train the baseline and the best model on the full training set, and score both once on validation.

	model	weighted_f1	macro_f1
0	Baseline (03 notebook)	0.8313	0.8192
1	Optimised	0.8436	0.8330

Analysis

- The optimised model reaches 0.8436 weighted F1 on validation, against 0.8313 for the baseline (+1.2 points).
- Macro F1 goes from 0.8192 to 0.8330.
- The gain comes from using all features. `binary=True` and `C=1` were already used in the baseline.

Conclusion

Final text model: TF-IDF on words and word pairs (all features, binary) with LinearSVC (C=1). Weighted F1 on validation: 0.8436.